

The Crypto Fear Channel: Asymmetric, Tail-Concentrated, and Regime-Dependent Volatility Spillover from Bitcoin to Equity Markets

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Abstract

Using daily returns on SPY, BTC-USD, and VIX from 2015-02 to 2026-04 ($N = 2,812$), we document three stylized facts about the crypto-to-equity volatility spillover. First, the spillover is *asymmetric*: Bitcoin downside volatility Granger-causes VIX at lags 1–5, whereas Bitcoin upside volatility does not (symmetric Granger tests covering lags 1–10 corroborate the broader directional result). Second, the spillover is *tail-concentrated with sign reversal*: the quantile-regression coefficient of VIX on BTC realized variance is significantly *negative* at lower quantiles (-2.78 at $\tau = 0.05$; -2.12 at $\tau = 0.25$), indicating bull-market decoupling, then turns positive at the median ($+2.82$ at $\tau = 0.5$), continues climbing through the upper interquartile range ($+9.39$ at $\tau = 0.75$), and amplifies to $+19.85$ at the 95th percentile, yielding a $7.0\times$ upper-tail amplification relative to the median. Third, the spillover is *regime-dependent*: in a five-subperiod breakdown, only 2020 survives Bonferroni correction ($F = 12.31$, $p < 10^{-6}$); 2015–2017 is marginally significant at the uncorrected 5% level ($p = 0.014$) and the remaining three subperiods are clearly non-significant. In the Diebold-Yilmaz framework, Bitcoin is a *net receiver*, rehabilitating the narrative from “crypto as a fear originator” to “crypto as a fear amplifier” conditional on pre-existing equity stress. Out-of-sample forecasts, however, show that adding BTC realized volatility does not statistically improve an AR model of VIX (Diebold-Mariano $t = -1.14$, $p = 0.26$), highlighting that Granger causality is a necessary but not sufficient condition for forecastability. Taken together, our findings reframe the crypto-equity spillover as a *crisis-time amplifier*: informative for ex-post risk attribution and margin design, but inadequate as an ex-ante predictor of equity fear.

Keywords: volatility spillover, Bitcoin, VIX, asymmetric causality, quantile regression, Diebold-Yilmaz, crypto-equity linkage

JEL: G11, G15, G17, C22

1 Introduction

Since the institutionalization of Bitcoin trading and the 2024 approval of U.S. spot BTC ETFs, the question of whether cryptocurrency markets transmit fear to traditional equity markets has acquired first-order practical importance for portfolio risk management, prudential supervision, and margin design. Yet the empirical literature has yielded an uneasy consensus. On one hand, correlation-based studies report rising BTC-equity co-movement in stress periods, suggesting that cryptocurrency is no longer a standalone asset class.¹ On the other hand,

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¹See, e.g., Bouri et al. (2020), Corbet et al. (2018), and Matkovskyy and Jalan (2019) for the early post-2017 literature.

return-predictability tests often fail to find that Bitcoin-based signals improve forecasts of equity volatility at practical horizons. This tension motivates a central question: is the crypto-equity fear channel *real but nonlinear*, *real but regime-dependent*, or an artifact of tail-period correlation without predictive content?

We resolve this tension by decomposing the spillover along three dimensions that the prior literature has typically examined in isolation: directional asymmetry, tail dependence, and regime stability. Each of these dimensions yields a stylized fact that materially changes how one should interpret correlation-based evidence.

Asymmetry. Using the asymmetric Granger-causality framework of Hatemi-J (2012), we separate Bitcoin positive and negative realized volatility and test whether each Granger-causes VIX at lags 1 through 5. We find that the negative branch carries the entire spillover: Bitcoin downside volatility Granger-causes VIX at every lag tested (1–5), while the upside branch is uniformly non-significant across the same lag window. This is not merely a statistical artifact of a leverage effect — the asymmetry is corroborated by standard symmetric Granger tests over the broader 1 through 10 lag range, where BTC-to-VIX causality is significant at lags 2–10. Our interpretation is behavioral: retail-heavy crypto markets transmit *fear*, not *euphoria*, into the institutional equity complex.

Tail concentration with sign reversal. Quantile regressions of VIX on BTC realized variance reveal a richer structure than a simple tail amplification: the slope *reverses sign* across the VIX distribution. At the lower quantiles the coefficient is significantly *negative* (-2.78 at $\tau = 0.05$; -2.12 at $\tau = 0.25$), consistent with a bull-market coexistence regime in which rising Bitcoin volatility coincides with *compressing* equity fear. The slope turns positive at the median ($+2.82$ at $\tau = 0.5$) and amplifies sharply in the upper tail ($+19.85$ at $\tau = 0.95$), an approximately $7.0\times$ amplification of the upper tail relative to the median ($\tau = 0.95/\tau = 0.5 \approx 7.04$). The spillover is therefore not a uniform channel, nor even a simple “flat-then-spikes” tail phenomenon, but a regime with a *directional switch* at the median of VIX: BTC volatility dampens equity fear in calm markets and amplifies it in panic markets. Conventional conditional-mean summaries and pooled correlation diagnostics — which collapse this sign-reversing heterogeneity into a single slope — systematically mischaracterize both the bull-market decoupling and the crisis-time amplification of crypto volatility on equity fear.

Regime dependence. A five-subperiod breakdown reveals that Granger significance is concentrated in 2020 after multiple-testing correction ($F = 12.31$, $p < 10^{-6}$), with 2015–2017 only marginally significant at the uncorrected 5% level ($p = 0.014$). Combined with the finding from Diebold and Yilmaz (2012) spillover-index analysis that Bitcoin is a *net receiver* rather than net sender, this suggests that the spillover is not a permanent feature of cross-market architecture but a *conditional amplification mechanism* that activates mainly during pre-existing equity stress.

The forecastability gap. A natural next question is whether these three stylized facts translate into out-of-sample predictive power. They do not. Comparing an $AR(p)$ specification of VIX against one augmented with BTC realized volatility over an OOS window of 1,826 days (2019–2026), the Diebold-Mariano test returns $t = -1.14$, $p = 0.26$ — statistically indistinguishable from zero under conventional inference. This result echoes a broader methodological point: *Granger causality is a necessary but not sufficient condition for forecastability*. An informative signal that appears only in tail events and only during rare regimes may be too sparse to outperform a well-specified autoregressive benchmark on average over a long OOS sample.

We make three contributions to the literature. First, we document a new empirical pattern in the BTC-to-VIX channel: the conditional response of equity fear to Bitcoin realized variance

reverses sign between the lower and upper halves of the VIX distribution and amplifies by approximately $7.0\times$ from the median to the 95th percentile. The sign reversal — statistically significant in both halves with bootstrap confidence intervals excluding zero at every quantile tested — is invisible to conditional-mean estimators and has not, to our knowledge, been documented in the cryptocurrency-equity spillover literature. Second, the COVID-2020 sub-period emerges as the only window that survives Bonferroni multiple-testing correction across five non-overlapping subperiods, while a -74 percentage-point net Diebold-Yilmaz directional spillover identifies Bitcoin as a fear *amplifier* (recipient), not originator (sender). Third, we pair each in-sample finding with an out-of-sample evaluation under a conventional Diebold-Mariano comparison, and the resulting OOS null ($t = -1.14$, $p = 0.26$) supplies a useful counterweight against promoting in-sample structure into unsupported forecasting claims (Conrad and Kleen, 2020). Each contribution rests on the four methodological building blocks — asymmetric Granger causality, quantile regression for tail dependence, rolling VAR spillover decomposition, and Diebold-Mariano forecast comparison — developed in §4; the novelty lies in their joint application to the BTC-equity fear channel and in the empirical patterns they collectively reveal.

The remainder of the paper is organized as follows. Section 2 surveys the crypto-equity spillover literature and positions our contribution. Section 3 describes the data and preliminaries. Section 4 details the four core methodological building blocks (asymmetric Granger, quantile regression, Diebold-Yilmaz, Diebold-Mariano), preceded by a symmetric-Granger baseline. Sections 5 and 6 present the main results and robustness checks. Section 7 reports the out-of-sample forecasting evaluation. Section 8 discusses mechanisms and policy implications. Section 9 concludes.

2 Literature Review

We organize the related literature along three threads: (i) the empirical literature on crypto-equity volatility spillover and the safe-haven debate, (ii) the methodological literature underlying our four building blocks (asymmetric Granger, quantile regression, Diebold-Yilmaz, Diebold-Mariano), and (iii) the recent calls for honest out-of-sample evaluation in volatility forecasting. Each thread has progressed independently; our contribution is to combine them within a single framework that decomposes the spillover into asymmetry, tail concentration, regime conditionality, and predictive power.

2.1 Crypto-equity volatility spillover and the safe-haven debate

The empirical literature on Bitcoin’s role in equity portfolios has evolved through three phases. The first phase, exemplified by Corbet et al. (2018) and Matkovskyy and Jalan (2019), documented that cryptocurrency price movements respond meaningfully to macro-financial news and that contagion-style co-movement with equity markets emerges in stress periods, contradicting the early “digital gold” framing. The second phase, motivated by COVID-19 and reviewed in Bouri et al. (2020), Conlon and McGee (2020), Shahzad et al. (2019), and Klein et al. (2018), sharpened the safe-haven question with mixed verdicts: wavelet- and DCC-based correlation analysis showed that Bitcoin’s risk profile is closer to a high-volatility risky asset than to gold, and that its safe-haven properties are at best regime-dependent. The third phase, post-2020, has documented increasing institutional integration: Yarovaya et al. (2022) report that information transmission between cryptocurrency and traditional asset markets intensified during the pandemic, and Akyildirim et al. (2020) link implied-volatility shocks to cryptocurrency returns.

Three gaps motivate our analysis. First, most prior studies test *symmetric* causality or *symmetric* correlation, conflating the upside and downside volatility channels that we will show carry economically distinct information. Second, the tail dependence of the spillover is typically inferred indirectly from rolling DCC or wavelet coherence, rather than directly estimated

through quantile regression. Third, sample windows often end at 2020 or include 2020 as a single observation, leaving the post-COVID regime-stability question unaddressed — a gap that matters precisely because COVID-2020 will turn out to be the structural watershed of the spillover. Our analysis closes these gaps by replacing each symmetric or correlation-based summary with a direct, regime-aware, asymmetric counterpart, while extending the sample through April 2026.

2.2 Methodological building blocks

Our four building blocks each have a settled methodological literature. Asymmetric Granger causality, the technique behind our headline asymmetry result, was formalized by Hatemi-J (2012), who decompose each series into positive and negative cumulative innovations and test their respective Granger predictability. Earlier nonparametric extensions of Granger causality (Diks and Panchenko, 2006) and asymmetric volatility-spillover tests (Hong, 2001) provide complementary nonparametric diagnostic tools for asymmetric-transmission testing. Each of these methods was developed to address the specific concern that linear Granger tests can mask sign-dependent transmission — precisely the asymmetry the present paper documents in the BTC-to-VIX channel.

Quantile regression, our tool for tail dependence, originates with Koenker and Bassett (1978) and has subsequently been extended into systemic-risk applications. Most relevant to our setting is Adrian and Brunnermeier (2016)’s CoVaR framework, in which the conditional value-at-risk of one institution given another’s stress is estimated through quantile regression on returns. We adopt the quantile-regression framework rather than CoVaR *per se* because our object of interest is the conditional response of equity *volatility* (VIX), not the conditional value-at-risk of equity returns; but the underlying statistical engine is the same.

The Diebold-Yilmaz spillover index is now a canonical tool for measuring directional volatility transmission across asset classes. Its development spans Diebold and Yilmaz (2009), Diebold and Yilmaz (2012), and Diebold and Yilmaz (2014), and a recent IMF policy note (Iyer, 2022) applies the framework specifically to cryptocurrency-equity spillovers. In the archived implementation used by this draft, we fit a rolling VAR on the ordered system $\{VIX, RV_{\text{spy}}^{(20)}, RV_{\text{btc}}^{(20)}\}$, select lags by AIC within each 252-day window (capped at 5), and compute a 10-step orthogonalized Cholesky FEVD via `statsmodels`. We therefore interpret the resulting spillover shares as ordering-dependent connectedness diagnostics rather than as fully generalized FEVD estimates.

Finally, our forecasting evaluation rests on the Diebold and Mariano (1995) test of equal predictive accuracy with the small-sample adjustment of Harvey et al. (1997). We report the conventional two-sided p -value and treat the OOS comparison as a separate object from the in-sample discovery exercise, rather than importing factor-discovery thresholds into forecast-comparison inference.

2.3 Honest out-of-sample evaluation in volatility forecasting

A recurring tension in the volatility-forecasting literature is the gap between in-sample statistical significance and out-of-sample economic value. Conrad and Kleen (2020)’s GARCH-MIDAS application demonstrates that incorporating macro-economic information through long-run components can deliver in-sample fit improvements that fail to translate into forecast accuracy gains. The cumulative evidence reviewed there motivates an evaluation discipline in which a new predictor must clear both (i) in-sample significance and (ii) out-of-sample improvement under Diebold-Mariano before being promoted to a structural claim. The present paper adopts this discipline and reports a transparent out-of-sample null — BTC realized volatility fails to improve AR-VIX forecasts at $t = -1.14$ — alongside its in-sample asymmetry, tail-amplification, and regime-stability findings.

3 Data and Preliminaries

3.1 Sample and variables

Our sample consists of daily data on three series: the SPDR S&P 500 ETF (SPY), Bitcoin in U.S. dollar terms (BTC-USD), and the CBOE Volatility Index (VIX). The current reported estimates come from the archived `k1025_v2.py` experiment, which downloads all three series from Yahoo Finance via `yfinance` with `auto_adjust=True`. A paper-local CSV snapshot (`paper/crypto-fear-channel/data/spy_btc_usd_vix_2015-2026.csv`) is archived for future pinning, but the headline results in this draft have not yet been fully regenerated directly from that file; for that reason this manuscript remains a major-revision draft rather than a submission-ready replication package. The full sample spans 2015-02-02 through 2026-04-08, yielding $N = 2,812$ trading-day observations after aligning BTC to the SPY/VIX trading calendar (we drop weekend BTC observations because VIX is undefined). The out-of-sample window used in §4.5 begins 2019-01-01 and runs through 2026-04-08, providing 1,826 OOS observations — spanning the COVID-19 shock (2020), the late-cycle bull market (2021), the 2022 bear, and the post-ETF recovery (2023–2026).

3.2 Constructed variables

We work with three constructed series. In the current archived implementation, SPY returns are computed as simple daily percentage changes, while BTC returns are computed as log returns from adjusted closes: $r_t^{\text{SPY}} = P_t^{\text{SPY}}/P_{t-1}^{\text{SPY}} - 1$ and $r_t^{\text{BTC}} = \log(P_t^{\text{BTC}}/P_{t-1}^{\text{BTC}})$. Realized volatility is then constructed as a 20-day rolling standard deviation of those return series, annualized by $\sqrt{252}$: $\text{RV}_t^{(20)} = \text{std}_{[t-19,t]}(r) \cdot \sqrt{252}$, computed separately for SPY and BTC. The VIX is used in level (it is already an annualized measure, %). For all forward-looking specifications we lag explanatory variables by one day (`signal.shift(1)` convention) to enforce causal predictability and rule out look-ahead.

For the asymmetric Granger tests of §4.2 we follow Hatemi-J (2012)’s sign-decomposition approach with one adaptation: rather than test the cumulative-positive and cumulative-negative innovations of BTC realized volatility directly, we first decompose BTC *returns* into positive and negative components ($r_t^{\text{BTC},+} = \max(r_t^{\text{BTC}}, 0)$ and $r_t^{\text{BTC},-} = -\min(r_t^{\text{BTC}}, 0)$) and then construct directional realized-volatility series from each component using the same 20-day rolling quadratic-mean as in the symmetric specification: $\text{RV}_t^{\text{BTC},\pm,(20)} = \sqrt{\frac{1}{20} \sum_{j=t-19}^t (r_j^{\text{BTC},\pm})^2} \cdot \sqrt{252}$. We take first differences of these directional RV series to ensure stationarity in the Granger specification (the level series carry near-unit-root persistence in 20-day windows), and use the first-differenced series $\Delta \text{RV}_t^{\text{BTC},\pm,(20)}$ as the explanatory variables in the asymmetric Granger tests. This adaptation preserves the key identification of Hatemi-J (2012) — separate sign-conditional Granger tests — while allowing the volatility-channel interpretation that motivates our application.

3.3 Descriptive statistics and stationarity

Table 1 reports descriptive statistics for the four primary series. BTC daily returns exhibit mean 0.158%, daily standard deviation 3.79%, and excess kurtosis 11.83 with pronounced negative skewness (−0.74). SPY daily returns are an order of magnitude less volatile in standard deviation (1.12%) and exhibit a still-higher excess kurtosis (14.15) in this sample window, concentrated in a handful of COVID-2020 and 2022 crisis days; the two series’ tail thickness is thus of comparable order in this window, with SPY’s edge reflecting a small set of extreme equity outliers rather than a reversal of the well-documented unconditional ranking of BTC over equities. The VIX averages 18.4 with maximum 82.7 (March 2020) and minimum 9.1, spanning the

calm 2017–2019 vol regime, the COVID shock, the 2022 inflation regime, and the 2024–2026 normalization.

Table 1: Descriptive statistics for daily series, 2015-02-02 to 2026-04-08 ($N = 2,812$).

Series	Mean	Std. dev.	Skewness	Excess kurtosis
$r^{\text{btc}} (\%)$	0.158	3.790	−0.739	11.826
$r^{\text{spy}} (\%)$	0.060	1.120	−0.307	14.150
VIX	18.382	7.110	—	—
$\text{RV}_{\text{btc}}^{(20)}$	0.543	0.266	—	—

VIX min/max are 9.14 / 82.69. $\text{RV}_{\text{btc}}^{(20)}$ min/max are 0.098 / 1.960 (annualized). Source: `experiments/k1025/k1025_v2_results.json`. `.descriptive_statistics` (audit fix 2026-06-10: values aligned to the cited v2 source, replacing v1 numbers).

Augmented Dickey-Fuller (ADF) stationarity tests reject the unit-root null for all three modeling series at conventional levels: $\text{RV}_{\text{btc}}^{(20)}$ ($t = -5.10$, $p < 0.001$, lags 20), VIX ($t = -5.71$, $p < 10^{-6}$, lags 9), and $\text{RV}_{\text{spy}}^{(20)}$ ($t = -4.46$, $p < 0.001$, lags 20). Stationarity is a precondition for the Granger and Diebold-Yilmaz analyses below, both of which assume covariance stationarity. We do not differentiate further; the analyses use the level series.

4 Methodology

We employ four methodological building blocks. The first three (§4.2–4.4) characterize *in-sample* cross-asset structure along three dimensions: directional asymmetry, tail dependence, and regime stability. The fourth (§4.5) tests *out-of-sample* predictive content.

4.1 Symmetric Granger causality (baseline)

As a baseline we test whether BTC realized volatility Granger-causes VIX in a standard linear specification with lags $\ell = 1, \dots, 10$. For each lag ℓ we estimate the unrestricted regression

$$\text{VIX}_t = \alpha_0 + \sum_{j=1}^{\ell} \alpha_j \text{VIX}_{t-j} + \sum_{j=1}^{\ell} \beta_j \text{RV}_{\text{btc}, t-j}^{(20)} + \varepsilon_t \quad (1)$$

and the restricted regression that drops the β_j terms. The joint null $H_0 : \beta_1 = \dots = \beta_{\ell} = 0$ is tested with the standard OLS F -test returned by `statsmodels.tsa.stattools.grangercausalitytests`. We therefore interpret these as plain lag-augmentation Granger diagnostics rather than as HAC-adjusted Wald tests. We tabulate F -statistics and p -values across lags 1 through 10 and treat the symmetric Granger result as the linear-causality benchmark against which the asymmetric and tail-conditional specifications below are compared.

4.2 Asymmetric Granger causality (Hatemi-J)

Following the sign-decomposition approach of Hatemi-J (2012), we test whether the directional RV series defined in §3.2 (the first-differenced rolling-RV series constructed separately from positive-only and negative-only BTC daily returns) carry distinct Granger predictive content for VIX. The asymmetric Granger specifications are estimated as two separate regressions, one

per branch:

$$\begin{aligned} \text{VIX}_t &= \alpha_0^+ + \sum_{j=1}^{\ell} \alpha_j^+ \text{VIX}_{t-j} + \sum_{j=1}^{\ell} \beta_j^+ \Delta \text{RV}_{t-j}^{\text{btc},+, (20)} + \varepsilon_t^+, \\ \text{VIX}_t &= \alpha_0^- + \sum_{j=1}^{\ell} \alpha_j^- \text{VIX}_{t-j} + \sum_{j=1}^{\ell} \beta_j^- \Delta \text{RV}_{t-j}^{\text{btc},-, (20)} + \varepsilon_t^-. \end{aligned} \quad (2)$$

We test the two null hypotheses, $H_0^+ : \beta_1^+ = \dots = \beta_{\ell}^+ = 0$ (BTC upside RV innovations do not Granger-cause VIX) and $H_0^- : \beta_1^- = \dots = \beta_{\ell}^- = 0$ (BTC downside RV innovations do not Granger-cause VIX), each with a standard F -test against the corresponding restricted regression. The Hatemi-J framework is appropriate when the suspected transmission channel is sign-dependent — our prior, motivated by retail-driven crypto trading and the asymmetric leverage effect in equity volatility, is precisely such asymmetry. We test lags $\ell \in \{1, 2, 3, 4, 5\}$.²

4.3 Quantile regression for tail dependence

Linear regression measures the conditional mean response of VIX to BTC RV; conditional on the documented heavy tails of both series and the sign-reversing structure we will document, the conditional mean is uninformative about regime-specific transmission. We follow Koenker and Bassett (1978) and estimate quantile regressions of VIX on BTC realized variance at five representative quantiles $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$:

$$Q_{\tau}(\text{VIX}_t \mid \text{RV}_{\text{btc}, t-1}^{(20)}) = \alpha_{\tau} + \beta_{\tau} \cdot \text{RV}_{\text{btc}, t-1}^{(20)}, \quad (3)$$

estimated via the standard Koenker-Bassett check-loss minimization. Standard errors and confidence intervals are obtained via 1,000-replication bootstrap of the regression sample. The tail-dependence ratio reported in our headline result is the upper-tail amplification $\beta_{0.95}/\beta_{0.50}$, which is approximately 7.04 in our sample. The conceptual link to Adrian and Brunnermeier (2016)’s CoVaR framework is direct: both rely on quantile regression to recover state-dependent cross-asset transmission, with the present paper applying the technique to volatility (VIX) rather than tail risk.

4.4 Diebold-Yilmaz spillover index

To measure directional volatility transmission, we estimate a rolling VAR on the 3-variable system $\{\text{VIX}, \text{RV}_{\text{spy}}^{(20)}, \text{RV}_{\text{btc}}^{(20)}\}$ and compute a 10-step orthogonalized forecast-error variance decomposition (FEVD) within each window. In the archived implementation, the lag order is selected by AIC in each rolling window (capped at 5 for stability), and the FEVD is generated by `statsmodels` using the Cholesky ordering shown above. The directional spillover from variable i to variable j is therefore an ordering-dependent share of variable j ’s forecast-error variance attributable to shocks in variable i . We report the total spillover index, the directional “to” and “from” indices, and BTC’s net spillover (to minus from) across rolling 252-trading-day windows. This should be read as a rolling connectedness diagnostic rather than as a fully generalized FEVD implementation.

4.5 Out-of-sample forecasting framework

We test whether the in-sample asymmetry, tail-amplification, and regime-stability findings translate into out-of-sample predictive power for VIX. The base forecast is an autoregressive

²The lag window 1–5 is chosen to match the timescale at which retail-investor sentiment plausibly transmits across asset markets; the symmetric Granger test in §4.1 extends to $\ell = 10$ to confirm robustness across longer horizons.

AR(p) model with lag length selected by AIC on the in-sample window (2015-02 to 2018-12); the augmented forecast adds lagged BTC realized volatility:

$$\text{VIX}_t = \alpha_0 + \sum_{j=1}^p \alpha_j \text{VIX}_{t-j} + \gamma \cdot \text{RV}_{\text{btc}, t-1}^{(20)} + u_t. \quad (4)$$

Both models are re-estimated on a rolling basis, and one-day-ahead point forecasts are generated for each day in the OOS window 2019-01-01 to 2026-04-08 (1,826 observations). Forecast errors are evaluated under squared-error loss, and the difference in loss between the two specifications is tested with the Diebold and Mariano (1995) test of equal predictive accuracy with the small-sample adjustment of Harvey et al. (1997). Failure to reject the null is interpreted as a transparent signal that, despite in-sample causal structure, BTC realized volatility does not improve VIX point forecasts on average over the long OOS sample.

5 Main Results

This section presents the three core empirical findings: directional asymmetry of the BTC-to-VIX channel (§5.1), tail concentration with sign reversal in the conditional response (§5.2), and regime-dependence of the Granger structure (§5.3).

5.1 Asymmetric Granger causality: only BTC downside transmits

Table 2 reports the central asymmetric-Granger result. The negative branch ($\Delta \text{RV}^{\text{btc}, -}$, the first-differenced downside directional RV series) Granger-causes VIX at every tested lag from 1 through 5, with F -statistics ranging from 8.27 to 21.78 and p -values uniformly below 10^{-5} . The positive branch ($\Delta \text{RV}^{\text{btc}, +}$) is not significant at any lag, with F -statistics between 0.17 and 2.04 and p -values between 0.15 and 0.95. The asymmetry is monolithic: there is no overlap region in which both branches contribute — the channel is fully one-sided.

Table 2: Asymmetric Granger causality: BTC realized-volatility innovations to VIX, lags 1–5.

Lag	BTC ⁺ → VIX		BTC [−] → VIX	
	F	p	F	p
1	2.04	0.153	21.78	3.2×10^{-6}
2	0.21	0.812	19.57	$< 10^{-8}$
3	0.20	0.896	13.11	1.7×10^{-8}
4	0.17	0.951	9.49	1.3×10^{-7}
5	0.29	0.919	8.27	$< 10^{-6}$

F -tests of the joint null that the lagged $\text{BTC}^{\pm}$ coefficients are zero in Eq. (2). The negative-branch test rejects the null at every lag with $p < 10^{-5}$; the positive branch fails to reject at any lag. Source: `experiments/k1025/k1025_v2_results.json` .`asymmetric_granger`.

The asymmetry is corroborated by symmetric Granger tests over a broader lag window. Pooling the positive and negative innovations into a single specification, the symmetric $\text{BTC RV} \rightarrow \text{VIX}$ test rejects the joint null at lags 2 through 10 with p -values from 4.9×10^{-3} (lag 2) down to below 10^{-6} (lags 3–10); the lag-1 specification is non-significant ($F = 0.59$, $p = 0.44$).

Asymmetric vs. symmetric Granger: reading the lag-1 result. The lag-1 contrast is the cleanest diagnostic in our test battery for why asymmetric decomposition matters. The symmetric specification at lag 1 averages over the (small, possibly zero) upside coefficient and the

(large, positive) downside coefficient, producing the pooled $F = 0.59$ ($p = 0.44$) non-rejection. Once the two branches are estimated separately as in Eq. (2), the downside-branch test recovers $F = 21.78$ ($p < 10^{-5}$): the lag-1 BTC-to-VIX channel is real but mechanically invisible to a sign-pooled symmetric test. Reading the two tables jointly: BTC RV does Granger-cause VIX, but the entire signal lives in the downside half of the BTC innovation distribution. This is a clean illustration of why asymmetric and symmetric Granger are not interchangeable diagnostics; using only the symmetric test would have led us to mis-time the predictive horizon by one lag.

5.2 Tail dependence with sign reversal

Table 3 reports the quantile-regression slope of VIX on lagged BTC realized variance across five representative quantiles. The most striking pattern is the *sign reversal* of the coefficient as the conditional VIX quantile rises: at $\tau = 0.05$ and $\tau = 0.25$ the slope is significantly *negative* ($\beta_{0.05} = -2.78$, $t = -13.84$; $\beta_{0.25} = -2.12$, $t = -4.89$), indicating that in the lower (calm) tail of the VIX distribution, higher BTC realized variance coincides with *lower* expected VIX. The slope crosses zero between $\tau = 0.25$ and $\tau = 0.5$ and turns sharply positive: $\beta_{0.5} = 2.82$ ($t = 4.15$), $\beta_{0.75} = 9.39$ ($t = 6.61$), $\beta_{0.95} = 19.85$ ($t = 8.17$).

Table 3: Quantile regression of VIX on lagged BTC realized variance, $N = 2,812$.

τ	0.05	0.25	0.50	0.75	0.95
β_τ	-2.78	-2.12	+2.82	+9.39	+19.85
SE (boot)	0.201	0.433	0.679	1.420	2.430
t -stat	-13.84	-4.89	4.15	6.61	8.17
95% boot CI	$[-3.32, -2.57]$	$[-2.88, -1.28]$	$[+1.46, +4.20]$	$[+6.53, +12.43]$	$[+14.36, +23.21]$

Slope coefficients from quantile regressions of VIX on lagged BTC realized variance; standard errors and t -statistics are based on 1,000 bootstrap replications, and the confidence-interval row reports the corresponding percentile bootstrap intervals. The sign of the slope reverses between the lower quantile and the median; the upper-tail amplification ratio is $\beta_{0.95}/\beta_{0.5} \approx 7.04$. Source: `experiments/k1025/k1025_v2_results.json.quantile_regression`.

The economically meaningful summary statistic is the upper-tail amplification ratio $\beta_{0.95}/\beta_{0.5} \approx 7.04$: the conditional sensitivity of VIX to BTC realized variance is roughly 7 times stronger at the 95th percentile of VIX than at the median. The sign reversal at lower quantiles is, to our knowledge, a new finding for crypto-equity spillover: it implies that pooled conditional-mean estimators average over a regime in which BTC volatility *dampens* expected equity fear (the bull-market decoupling we discuss in §8) and a regime in which it *amplifies* fear, producing a pooled slope that mischaracterizes both halves.

5.3 Regime dependence: COVID-2020 as structural watershed

We split the 2015–2026 sample into five non-overlapping subperiods — pre-mania (2015–2017), crypto-winter (2018–2019), COVID (2020), bull-bear (2021–2022), and recovery-plus-ETF (2023–2026) — and re-estimate the BTC-to-VIX Granger test within each subperiod with AIC-selected lag length. Table 4 reports the result. Granger causality survives the Bonferroni multiple-testing correction ($p < 0.01/5$) only during 2020 ($F = 12.31$, $p = 1.6 \times 10^{-7}$, lag 3); among the other four subperiods, 2015–2017 is marginally significant at the uncorrected 5% level ($F = 2.88$, $p = 0.014$, Table 4) but does not pass the corrected threshold, and the remaining three fail to reject at any conventional level.

Table 4: Subperiod Granger tests of BTC realized volatility on VIX, AIC-best lag.

Subperiod	N	Best lag	F	p
2015–2017 (Pre-mania)	735	5	2.88	0.014
2018–2019 (Crypto winter)	503	1	0.37	0.544
2020 (COVID)	253	3	12.31	1.6×10^{-7}
2021–2022 (Bull-Bear)	503	1	2.03	0.155
2023–2026 (Recovery+ETF)	818	2	0.16	0.850

Granger F -tests within each subperiod with AIC-selected lag length. Only 2020 rejects the non-causality null (Bonferroni threshold also cleared). Source: `experiments/k1025/k1025_v2_results.json`
`.subperiod_granger`.

The EWMA dynamic correlation (RiskMetrics, $\lambda = 0.94$) by VIX regime corroborates the regime-dependence finding from a different angle. Sorting trading days into four VIX regimes (Low: $VIX \leq 15$; Normal: $15 < VIX \leq 25$; High: $25 < VIX \leq 35$; Crisis: $VIX > 35$), the mean BTC-SPY EWMA dynamic correlation (RiskMetrics, $\lambda = 0.94$) rises near-monotonically from 0.07 in the Low regime to 0.27 in Normal, peaking at 0.45 in High, and remaining elevated at 0.41 in Crisis (Table 5; the slight Crisis dip from 0.45 to 0.41 reflects the small Crisis-regime sample of $n = 63$). The percentage of days with positive correlation rises from 62% in Low to 95% in High and 98% in Crisis. Bitcoin’s purported diversification value relative to equities is therefore essentially absent precisely when investors most need it.

Table 5: BTC-SPY EWMA dynamic correlation (RiskMetrics, $\lambda = 0.94$) by VIX regime, $N = 2,812$ days total.

VIX regime	N	Mean	Median	Std. dev.	% positive
Low ($VIX \leq 15$)	1,037	0.068	0.053	0.190	61.9
Normal (15–25)	1,386	0.265	0.305	0.260	82.7
High (25–35)	326	0.449	0.480	0.206	95.1
Crisis ($VIX > 35$)	63	0.409	0.488	0.162	98.4

EWMA (RiskMetrics, $\lambda = 0.94$) BTC-SPY dynamic correlation aggregated within each VIX regime (audit fix 2026-06-10: regime cutpoints and correlation-estimator label aligned to `k1025_v2.py` implementation; an earlier draft labelled this DCC and used 22/30 cutpoints). The High and Crisis regime correlations contradict the early “digital gold” thesis and corroborate the crisis-time amplification documented in Tables 2–3. Source: `experiments/k1025/k1025_v2_results.json` `.dcc_correlation_by_regime`.

Finally, the Diebold-Yilmaz-style spillover analysis quantifies directionality. Across the 512 rolling 252-day windows in the sample, the total spillover index averages 90.1% with very low variation (standard deviation 0.22 percentage points; min 89.73%, max 90.84%), indicating that volatility shocks among $\{VIX, RV_{\text{spy}}^{(20)}, RV_{\text{btc}}^{(20)}\}$ are persistently and almost completely cross-transmitted. The decomposition reveals, however, that BTC is a *net receiver*: the average shock contribution *from* BTC to the system is 23.7%, while BTC’s average net spillover is -74.4 percentage points. Combined with the Granger and quantile-regression results, the directional finding reframes the crypto-equity link from “crypto as fear originator” to “crypto as fear amplifier conditional on pre-existing equity stress” — a refinement that has direct implications for prudential policy (§8).

6 Robustness

This section reports the one robustness exercise that is currently archived in a traceable way for the active manuscript: the time-series stability of the rolling spillover index. Other robustness ideas considered in earlier drafts remain prospective until they are written back into the experiment outputs.

6.1 Spillover-index stability across rolling windows

The Diebold-Yilmaz-style total spillover index is computed on each of 512 rolling 252-trading-day windows. The mean is 90.11% with a standard deviation of only 0.22 percentage points (i.e., the rolling spillover ranges from 89.73% to 90.84%, a peak-to-trough swing of about 1.11 percentage points across the entire sample). The remarkable stability of the total index — a swing of just 1.0 percentage point across an 11-year period that includes COVID-2020, the 2022 inflation regime, and the 2024 ETF approval — indicates that the *cross-asset variance-decomposition geometry* of the BTC–SPY–VIX system is essentially constant, even as individual pairwise correlations and Granger structures shift dramatically. The corollary is that the regime-dependence we document in §5.3 is *not* an artifact of structural breaks in the underlying VAR system: the variance-decomposition shares move by less than 1 percentage point even as the within-period Granger F -statistic ranges from a minimum of 0.16 (2023–2026 recovery+ETF) to a maximum of 12.31 (COVID-2020) — a 48-fold spread. The spillover index and the Granger test are measuring different objects — aggregate connectedness vs. directed predictability — and the regime story belongs to the latter.

The directional decomposition is similarly stable. BTC’s average “from-system” spillover (the share of BTC RV variance attributable to VIX and SPY shocks) is 23.7% across the rolling windows; the average net spillover (to minus from) is -74.4 percentage points. BTC remains a net receiver in every window: in no rolling 252-day subsample does the variance-decomposition flip BTC into a net-sender position. The “crypto as fear amplifier conditional on equity stress” interpretation is therefore a structural feature of this particular rolling-VAR decomposition, not a window-specific finding.

Deferred robustness work. Three additional exercises considered in earlier drafts are intentionally not carried as active evidence here: a uniform-lag re-estimation of the subperiod Granger tests, a dedicated pre-/post-ETF split, and a cross-asset VIXN/QQQ replication. The first two are not currently written back into the experiment JSON that underlies this manuscript, and the third still relies on an older, uncorrected methodology rather than the corrected v2 specification. They remain natural next steps, but they are not treated as completed robustness results in this draft.

7 Out-of-Sample Forecasting Evaluation

The three in-sample results in §5 establish that the BTC-to-VIX channel is asymmetric, tail-concentrated with sign reversal, and concentrated in the COVID-2020 subperiod. The natural next question is whether this in-sample structure translates into out-of-sample forecasting power. We test the augmented specification of Eq. (4) against the AR baseline over the 1,826-day OOS window 2019-01-01 to 2026-04-08.

The Diebold-Mariano test under squared-error loss returns $t = -1.14$ with $p = 0.26$, both well below the Harvey et al. (2016) $|t| > 3$ threshold for newly proposed predictors and even below the conventional $|t| > 1.96$ benchmark. We return to the methodological reconciliation of this OOS null with the strong in-sample asymmetric and tail-conditional structure documented

in §5 in the discussion of Granger causality versus forecastability (§8.2).³The augmented specification’s MSE is 4.662 versus the AR baseline’s 4.649 — a 0.28% *deterioration* rather than improvement. Under MAE, the augmented specification deteriorates by 0.43%; under QLIKE, the two specifications are statistically indistinguishable to four decimals.

Table 6: Out-of-sample forecast evaluation: AR baseline vs. AR+BTC RV augmented, OOS 2019-01-01 to 2026-04-08 ($N = 1,826$).

Loss	AR baseline	AR+BTC RV	Δ (%)	DM t -stat
MSE	4.649	4.662	−0.28	−1.14
MAE	1.258	1.263	−0.43	—
QLIKE	6.923	6.923	≈ 0	—

Δ is the percentage improvement of the augmented model relative to the AR baseline; negative values indicate deterioration. The Diebold-Mariano t -statistic is computed under squared-error loss with HAC standard errors per Diebold and Mariano (1995), with the small-sample adjustment of Harvey et al. (1997). The reported two-sided p -value is 0.26, so the augmented model does not beat the AR baseline under conventional inference. Source: `experiments/k1025/k1025_v2_results.json .forecast_evaluation`.

Subsample by VIX regime. A natural concern is that the pooled OOS test obscures crisis-period predictability that the asymmetric Granger and quantile-regression results highlight. We split the 1,826-day OOS window into a Crisis subsample ($VIX > 25$, $N = 347$) and a Normal subsample ($VIX \leq 25$, $N = 1,479$) and re-run the DM test within each. Neither subsample rejects predictive equality: the Crisis subsample yields DM $t = -0.22$ ($p > 0.80$) with the augmented model deteriorating by 0.10%, and the Normal subsample yields DM $t = -1.96$ ($p \approx 0.05$) with the augmented model deteriorating by 0.70%. The crisis-period subsample is too small to identify modest differences in average point-forecast loss with precision — and the in-sample evidence does not anyway predict that the augmented model would dominate within the crisis subsample on *point-forecast* loss; the in-sample analysis predicts that the conditional *slope* of VIX on BTC RV is regime-dependent, which is a different statistical object than the conditional mean.

Interpretation. The forecasting null is robust across pooled and regime-stratified analysis. We interpret the result as a substantive contribution rather than as a power deficiency. Conrad and Kleen (2020)’s methodological argument is directly relevant: an in-sample slope that lives in the upper tail of the conditional VIX distribution and only during 1 of 5 subperiods is exactly the kind of sparse signal that fails to outperform a well-specified autoregression on average over a long OOS sample. The honest joint reporting of (i) significant asymmetric Granger with F -statistics up to 21.78, (ii) tail amplification of $7.0\times$, (iii) regime concentration in 2020, and (iv) failed OOS DM at $|t| < 1.5$ supplies a cleaner template for empirical claims about cross-asset predictors than the “in-sample-only” or “OOS-only” modes that dominate the existing crypto-equity spillover literature.

³The Online-replication archive reports the single rolling DM statistic and pooled MSE/MAE/QLIKE losses (Table 6); diagnostics on the rolling-window in-sample $\hat{\gamma}$ path (sign stability, t -statistic distribution across re-estimations) are left to a follow-up extension. The pooled DM null does not by itself rule out a stable, sign-conditional γ ; what we report is a discipline-level no-improvement result on average loss, not a structural rejection of the underlying γ .

8 Discussion

This section interprets the empirical findings, addresses the Granger-versus-forecastability tension, and draws policy implications.

8.1 Asymmetry mechanism: retail-driven downside transmission

The asymmetric Granger result (Table 2) shows that BTC downside volatility transmits to VIX while BTC upside volatility does not. This pattern is unlikely to reflect a symmetric leverage effect within Bitcoin’s own price process, because the leverage effect would generate the asymmetry within the BTC-VIX joint distribution but would not, on its own, prevent a positive BTC volatility shock from spilling over to expected equity fear. One candidate interpretation is behavioral and market-structure based: Bitcoin trading is materially more retail-driven than U.S. large-cap equity trading, and downside moves may carry a more salient risk signal than upside moves. Under that interpretation, negative BTC shocks could co-move with broader risk-off positioning more strongly than positive shocks do. We do not structurally identify this mechanism here, so the mechanism language should be read as suggestive rather than as established evidence of margin-cascade contagion.

8.2 Granger causality $\neq$ forecastability: a methodological lesson

The juxtaposition of the highly significant in-sample asymmetric Granger result (F up to 21.78, $p < 10^{-5}$) with the failed out-of-sample DM test ($t = -1.14$) is, on its surface, paradoxical. Two reconciliations are consistent with our evidence. First, the in-sample signal is concentrated in two thin slices of the joint distribution: the negative branch of BTC innovations (with no contribution from the positive branch, by Table 2), and the upper tail of the VIX distribution (with sign reversal in the lower tail, by Table 3). Combined, these two slices identify a sparse signal — BTC negative innovations $\rightarrow$ upper-quantile VIX — that produces a strong F -statistic on average over the sample but does not improve point forecasts of VIX on average over the OOS window, because most OOS days are not in the upper-tail VIX regime and most BTC innovations are not negative. Second, the regime-conditional finding (§5.3) localizes the signal to one of five subperiods (2020); even if the predictive content within 2020 is real, it is too brief to dominate the point-forecast loss averaged across 1,826 OOS days. The methodological lesson is that the Granger test and the DM test are answering different questions: “does X predict Y at *any* horizon, conditional on full information about the joint distribution?” versus “does augmenting an autoregressive forecast of Y with X reduce expected loss *on average* across an OOS sample?” Our evidence supports the former but not the latter.

8.3 Implications for prudential policy and margin design

The crisis-conditional spillover may have policy relevance, but the present evidence supports only a narrower statement than some earlier drafts suggested. Where systemic-risk frameworks for positive-feedback equity strategies focus on *intra-equity* concentration of correlated trading flow during stress, the present paper is consistent with an *external-market amplification* channel in which cryptocurrency volatility co-moves more strongly with equity fear during crisis than during calm periods. That is enough to suggest that supervisors and risk managers should not assume complete decoupling between crypto and equity stress states. It is *not* enough, by itself, to justify a numeric stress-test threshold, to infer causal margin-cascade mechanics, or to claim measurable retail-protection externalities. We therefore read the policy implication as conditional and qualitative: crisis-state crypto volatility belongs on the monitoring dashboard, while stronger structural claims require richer identification.

8.4 Limitations and future work

Three limitations qualify our findings. First, the COVID-2020 subperiod is the only window in which Granger causality clears conventional thresholds, and that subperiod includes both the March-2020 panic and the subsequent monetary-policy regime shift; disentangling pure-fear contagion from policy-induced volatility transmission is beyond our reduced-form framework and would require structural identification through high-frequency event studies. Second, our evidence is at the daily frequency; intraday transmission — where liquidation cascades might plausibly originate — is invisible to the daily Granger and quantile tests we deploy. Third, the current archived implementation still needs a cleaner end-to-end replication pass: the headline estimates come from the corrected v2 experiment, but data pinning, FEVD-ordering sensitivity, and cross-asset replication have not yet been rebuilt into a single symmetric manuscript package. Those replication tasks are the main prerequisite before stronger external claims should be made.

9 Conclusion

Using daily data on SPY, BTC-USD, and VIX from February 2015 through April 2026 ($N = 2,812$), we document three stylized facts about the crypto-to-equity volatility spillover. First, the spillover is *asymmetric*: Bitcoin downside volatility Granger-causes VIX at every tested lag from 1 through 5 (F between 8.27 and 21.78, p uniformly below 10^{-5}), while the upside branch is non-significant at all lags. Second, the spillover is *tail-concentrated with sign reversal*: the conditional slope of VIX on lagged BTC realized variance is significantly negative in the lower quantiles ($\beta_{0.05} = -2.78$), turns positive at the median ($\beta_{0.5} = +2.82$), and amplifies $7.0\times$ to $\beta_{0.95} = +19.85$ at the upper tail. Third, the spillover is *regime-dependent*: of five non-overlapping subperiods spanning 2015 through 2026, only the 2020 COVID window rejects non-causality at conventional levels ($F = 12.31$, $p < 10^{-6}$). In the Diebold-Yilmaz framework Bitcoin is a *net receiver* (mean net spillover -74.4 percentage points), reframing the crypto-equity link from “crypto as fear originator” to “crypto as fear amplifier conditional on pre-existing equity stress.”

The corresponding out-of-sample test, however, returns a transparent null: augmenting an AR baseline of VIX with lagged BTC realized volatility does not reduce expected forecast loss over the 1,826-day OOS window 2019–2026 (DM $t = -1.14$, $p = 0.26$). We interpret this null as evidence that the in-sample signal is too sparse — localized in the BTC negative branch, the upper-quantile VIX regime, and the COVID subperiod — to dominate point-forecast loss averaged over the long OOS sample, and we treat the joint reporting of in-sample structure with out-of-sample discipline as a contribution in itself, illustrating that Granger causality and forecastability answer related but distinct questions.

The findings reframe the crypto-equity volatility link in three ways. Methodologically, asymmetric and tail-conditional estimators recover structure that pooled conditional-mean tools systematically obscure. Empirically, the COVID-2020 watershed implies that crypto-equity transmission is a crisis-time phenomenon rather than a steady-state channel, with dormant periods during which the two markets behave as approximately independent. Substantively, the paper is strongest as a reduced-form characterization of when the BTC–VIX link appears and when it vanishes; stronger claims about structural mechanism, cross-asset generality, or operational policy calibration remain future work. We see three natural extensions: high-frequency replication to identify the within-day transmission mechanism, a fully pinned rerun of the headline experiment, and a cross-asset rebuild under a methodologically symmetric specification. The asymmetric, tail-concentrated, regime-dependent crypto fear channel is, on the evidence, real — but real in a sense that requires conditioning on crisis, on tail, and on direction to be visible at all.

References

- Adrian, T. and Brunnermeier, M.K. (2016). CoVaR. *American Economic Review*, 106(7):1705–1741. <https://doi.org/10.1257/aer.20120555>
- Akyildirim, E., Corbet, S., Lucey, B., Sensoy, A., and Yarovaya, L. (2020). The relationship between implied volatility and cryptocurrency returns. *Finance Research Letters*, 33:101212. <https://doi.org/10.1016/j.frl.2019.06.010>
- Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3):817–858. <https://doi.org/10.2307/2938229>
- Bouri, E., Shahzad, S.J.H., Roubaud, D., Kristoufek, L., and Lucey, B. (2020). Bitcoin, gold, and commodities as safe havens for stocks: New insight through wavelet analysis. *The Quarterly Review of Economics and Finance*, 77:156–164. <https://doi.org/10.1016/j.qref.2020.03.004>
- Conlon, T. and McGee, R. (2020). Safe haven or risky hazard? Bitcoin during the Covid-19 bear market. *Finance Research Letters*, 35:101607. <https://doi.org/10.1016/j.frl.2020.101607>
- Conrad, C. and Kleen, O. (2020). Two are better than one: Volatility forecasting using multiplicative component GARCH-MIDAS models. *Journal of Applied Econometrics*, 35(1):19–45. <https://doi.org/10.1002/jae.2742>
- Corbet, S., Meegan, A., Larkin, C., Lucey, B., and Yarovaya, L. (2018). Exploring the dynamic relationships between cryptocurrencies and other financial assets. *Economics Letters*, 165:28–34. <https://doi.org/10.1016/j.econlet.2018.01.004>
- Diebold, F.X. and Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263. <https://doi.org/10.1080/07350015.1995.10524599>
- Diebold, F.X. and Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. *The Economic Journal*, 119(534):158–171. <https://doi.org/10.1111/j.1468-0297.2008.02208.x>
- Diebold, F.X. and Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1):57–66. <https://doi.org/10.1016/j.ijforecast.2011.02.006>
- Diebold, F.X. and Yilmaz, K. (2014). On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134. <https://doi.org/10.1016/j.jeconom.2014.04.012>
- Diks, C. and Panchenko, V. (2006). A new statistic and practical guidelines for nonparametric Granger causality testing. *Journal of Economic Dynamics and Control*, 30(9–10):1647–1669. <https://doi.org/10.1016/j.jedc.2005.08.008>
- Harvey, D., Leybourne, S., and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291. [https://doi.org/10.1016/S0169-2070\(96\)00719-4](https://doi.org/10.1016/S0169-2070(96)00719-4)
- Harvey, C.R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68. <https://doi.org/10.1093/rfs/hhv059>
- Hatemi-J, A. (2012). Asymmetric causality tests with an application. *Empirical Economics*, 43(1):447–456. <https://doi.org/10.1007/s00181-011-0484-x>

- Hong, Y. (2001). A test for volatility spillover with application to exchange rates. *Journal of Econometrics*, 103(1–2):183–224. [https://doi.org/10.1016/S0304-4076\(01\)00043-4](https://doi.org/10.1016/S0304-4076(01)00043-4)
- Iyer, T. (2022). Cryptic connections: Spillovers between crypto and equity markets. *IMF Global Financial Stability Notes*, 2022/01. <https://doi.org/10.5089/9781616358068.065>
- Klein, T., Pham Thu, H., and Walther, T. (2018). Bitcoin is not the new gold — A comparison of volatility, correlation, and portfolio performance. *International Review of Financial Analysis*, 59:105–116. <https://doi.org/10.1016/j.irfa.2018.07.010>
- Koenker, R. and Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50. <https://doi.org/10.2307/1913643>
- Matkovskyy, R. and Jalan, A. (2019). From financial markets to Bitcoin markets: A fresh look at the contagion effect. *Finance Research Letters*, 31:93–97. <https://doi.org/10.1016/j.frl.2019.04.007>
- Shahzad, S.J.H., Bouri, E., Roubaud, D., Kristoufek, L., and Lucey, B. (2019). Is Bitcoin a better safe-haven investment than gold and commodities? *International Review of Financial Analysis*, 63:322–330. <https://doi.org/10.1016/j.irfa.2019.01.002>
- Yarovaya, L., Brzeszczyński, J., Goodell, J.W., Lucey, B., and Lau, C.K.M. (2022). Rethinking financial contagion: Information transmission mechanism during the COVID-19 pandemic. *Journal of International Financial Markets, Institutions and Money*, 79:101589. <https://doi.org/10.1016/j.intfin.2022.101589>